

Week 6 · Trace Worksheet — Deliverable 1

Instructions: Print this page landscape on A4. Fill in every cell. Use hex for all register values. Write register values **before** the cycle's update. Circle each row where ACC changes.

Program to trace:

```
0x000: CLOAD 10      (0x200A)    ACC ← 10
0x001: DSTORE 0x100 (0x4100)    mem[0x100] ← ACC
0x002: DLOAD  0x100 (0x3100)    ACC ← mem[0x100]
0x003: ADD     0x101 (0x7101)    ACC ← ACC + mem[0x101]
0x004: HALT                    (0x0000)    stop
```

Data: mem[0x100] = 0x0000, mem[0x101] = 0x0005

Cycle	State	PC	MAR	MBR	IR	ACC	Bus driver	Active loads/enables	Notes
1	fetch 1	0x000	—	—	—	0x0000	bus_pc	mar_ld	MAR ← PC
2	fetch 2	0x000	0x000	—	—	0x0000	bus_mem	mba_ld	MAR ← MAR PC ← PC + 1
3	' 3	0x001	0x000	0x200A	—	0x0000	bus_mbr	ir_ld	IR ← MBR
④	load 1	0x001	0x000	0x200A	0x200A	0x0000	bus_ir_se	acc_ld	ACC ← 0x000A
5	fetch 1	0x001	0x000	0x200A	0x200A	0x000A	bus_pc	mar_ld	MAR ← PC
6	' 2	0x001	0x001	0x200A	0x200A	' 1	bus_mem	mba_ld	MAR ← MAR PC ← PC + 1

←	← CLOAD done (cycle 4)								
7	fetch 3	0x002	0x001	0x4100	0x200A	r1	bus_mbr	pr_id	IRE-MBR
8	direct 1	0x002	0x001	0x4100	0x4100	r1	bus-br_lower	mem_id	MAP ← IR[11:0]
9	" 2	0x002	0x100	0x4100	0x4100	r1	bus_mem	mbz_id	MBR ← mem[MAP]
10	store 1	0x002	0x100	(state)	r1	r1	bus_acc	mem_wz	ACC ← mem[MAP] ← ACC
11	fetch 1	0x002	0x100	0x000A	r1	r1	bus_pc	mem_id	MAP ← PC
←	← DSTORE done (cycle 10)								
12	fetch 2	0x002	0x002	0x000A	r1	r1	bus_mem	mbz_id	MAP ← PC ← PC+1
13	" 3	0x003	0x002	0x3100	r1	r1	bus_mbr	pr_id	IRE-MBR
14	direct 1	0x003	0x002	0x3100	0x3100	r1	bus-br_lower	mem_id	MAP ← IR[11:0]
15	" 2	0x003	0x100	0x3100	r1	r1	bus_mem	mbz_id	MBR ← mem[MAP]
16	load 1	0x003	0x100	0x000A	r1	r1	bus_mbr	acc_id	ACC ← MBR
17	fetch 1	0x003	0x100	0x000A	r1	r1	bus_pc	mem_id	MAP ← PC
←	← DLOAD done (cycle 16)								
18	fetch 2	0x003	0x003	0x000A	r1	r1	bus_mem	mbz_id	MAP ← PC ← PC+1
19	" 3	0x004	0x003	0x7101	r1	r1	bus_mbr	pr_id	IRE-MBR
20	direct 1	0x004	0x003	0x7101	0x7101	r1	bus-br_lower	mem_id	MAP ← IR[11:0]
21	" 2	0x004	0x101	0x7101	r1	r1	bus_mem	mbz_id	MBR ← mem[MAP]
22	add 1	0x004	0x101	0x0005	r1	r1	bus_alu	acc_id	ACC ← ACC + ALU

23	fetch 1	0x004	0x1001	0x0005	'1'	0x00F	bus_pc	mar_ld	mar=0x004
←	← ADD done (cycle 22)								
24	fetch 2	0x004	0x004	0x0005	"	"	bus_mem	mbr_ld	more-mem PC=PC+1
25	" 3	0x005	0x004	0x0000	"	"	bus_mbr	ir_ld	IRE-MBR
26	halt 1	0x005	0x004	0x0000	0x0000	"	—	—	—
←	← HALT done (cycle 26) — done='1'								

Final values (fill in after completing the trace):

Register/Memory	Expected	Your answer	Match?
ACC	0x000F (15)	0x00f	✓
PC	0x005	0x005	✓
mem[0x100]	0x000A (10)	0x000A	✓
Total cycles	count yours	26	

State name reference (use these exact names): resetState · fetch1 · fetch2 · fetch3 · halt1 · neg1 · neg2 · cload1 · direct1 · direct2 · load1 · add1 · and1 · store1 · branch1 · indirect1 · indirect2 · iload1 · istore1 · brind1 · brind2

Bus driver reference: bus_pc · bus_mbr · bus_acc · bus_mem · bus_ir_lower · bus_ir_se · bus_alu

Load/enable reference: mar_ld · mbr_ld · ir_ld · acc_ld · pc_ld ·

pc_inc · mem_we